Foundational IT Skills

Experiments 1 & 2

Dept of Artificial Intelligence and Machine Learning

Today's Session

Experiment 1

Hardware & Software Exploration

Experiment 2

File and Folder Management in a Linux GUI

- Understand basic computer hardware
- Identify common hardware components and ports
- Distinguish system software and application software
- Understand Linux and GUI
- Create and manage files and folders

Syllabus Alignment

Module I

Basic Computer and Operating System Skills

- **Experiment 1:** Hardware & Software Exploration
- **Experiment 2:** File and Folder Management in a Linux GUI

Course Outcome

CO1: Identify the basic functions and features of Computers, Operating Systems, and Internet applications.

Experiment 1

Hardware & Software Exploration

What is a Computer?

Simple Definition

A computer is an electronic device that accepts data, processes it, stores it, and produces information.



Hardware and Software

Hardware

Physical parts of a computer that we can see and touch.

- Keyboard
- Mouse
- Monitor
- CPU
- RAM
- SSD / HDD

Software

Programs and instructions that tell the computer what to do.

- Ubuntu
- Windows
- Chrome
- VLC
- LibreOffice

Computer = Hardware + Software

Think About a Television

Physical TV	Programs inside TV
Can be seen and touched.	Control its functions.
Hardware	**Software**

Remember

Hardware needs software to perform useful tasks, and software needs hardware to run.

CPU – Central Processing Unit

Simple Definition

The CPU processes instructions and performs calculations.

- Often called the “brain” of the computer
- Executes instructions
- Processes data
- Controls computer operations

Example

If we enter:

$$25 + 15$$

the CPU processes the instruction and performs the calculation.

CPU and System Unit are Not the Same

Common Misconception

People often point to the computer cabinet and call it the "CPU".

- The cabinet is the **System Unit**
- The CPU/processor is a component inside it

System Unit

$$\Downarrow$$

CPU + RAM + Storage + Motherboard + Other Components

RAM – Random Access Memory

Simple Definition

RAM is temporary memory used by the computer while programs are running.

- Used for current/active tasks
- Fast working memory
- RAM is **volatile**
- Data is lost when power is switched off

Example

When Chrome, a calculator and a document are open, they use RAM while they are running.

RAM – A Simple Analogy

Study Table

Books currently being used.

Temporary Working Space

Cupboard

Books stored for future use.

Long-Term Storage

Study Table $\approx$ RAM
Cupboard $\approx$ Storage

RAM vs Storage

RAM	**Storage**
Temporary	Long-term
Volatile	Non-volatile
Used while programs run	Stores files and programs
Working memory	Secondary storage
Example: 8 GB RAM	Example: 512 GB SSD

Question

What happens to data in RAM when the computer is switched off?

HDD – Hard Disk Drive

Definition

HDD is a secondary storage device used to store the operating system, software, files and user data.

- Permanent storage
- Non-volatile
- Uses mechanical components

Examples of stored data

Documents Photos Videos Projects

SSD – Solid State Drive

Definition

SSD is a storage device that uses flash memory and has no moving parts.

- Faster access
- Lower power consumption
- More durable
- No moving mechanical parts

HDD vs SSD

HDD	SSD
Mechanical storage	Flash-memory storage
Moving parts	No moving parts
Generally slower	Generally faster
Less durable mechanically	More durable

Think

Two computers can both have 512 GB storage, but their performance can differ if one uses HDD and the other uses SSD.

Peripheral Ports

Simple Definition

A port is an interface through which another device can be connected to a computer.

- USB
- HDMI
- VGA
- Ethernet
- Audio
- USB Type-C
- Power port

Common Peripheral Ports

Port	Common Use
USB	Keyboard, mouse, pen drive
HDMI	Monitor / projector
VGA	Display connection
Ethernet	Wired network connection
Audio	Speakers / headphones
USB Type-C	Data transfer / charging
Power	Power supply

Classroom Activity: Identify the Port

Your Task

Look at the computer and identify the ports.

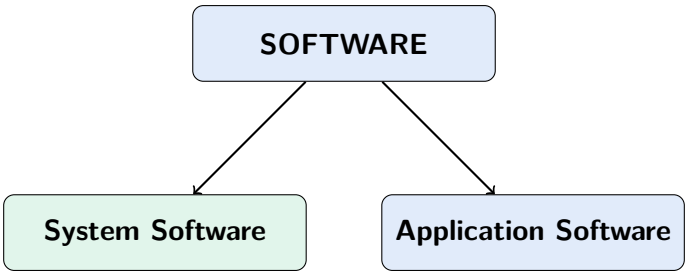
1. Which port is used for a pen drive?
2. Which port is commonly used for a projector?
3. Which port is used for a wired network?
4. Which port can be used for headphones?

Observe $\rightarrow$ **Identify** $\rightarrow$ **Explain**

Software

Definition

Software is a collection of programs and instructions that directs a computer to perform tasks.



System Software

Definition

System software controls the operation of the computer and manages hardware resources.

- Operating System
- Device Drivers
- Language Translators
- Utility Software

Examples

Ubuntu Windows Device Drivers

Application Software

Definition

Application software performs specific tasks for the user.

Application	**Purpose**
Word	Documents
Excel	Spreadsheets
PowerPoint	Presentations
Chrome	Web browsing
VLC	Media playback
Calculator	Calculations

System Software vs Application Software

System Software	Application Software
Manages computer resources	Performs user tasks
Provides the basic platform	Runs for a specific purpose
Ubuntu	Chrome
Windows	VLC
Device Driver	Calculator

Quick Classroom Quiz

System or Application?

1. Ubuntu
2. Google Chrome
3. VLC Media Player
4. Device Driver
5. Calculator
6. Operating System

Quick Classroom Quiz

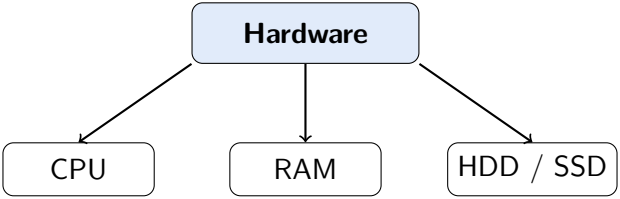
System or Application?

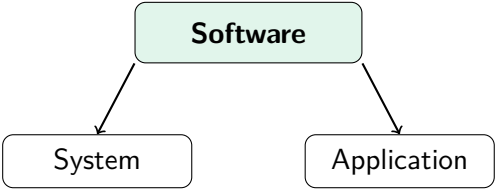
1. Ubuntu
2. Google Chrome
3. VLC Media Player
4. Device Driver
5. Calculator
6. Operating System

Answers:

Ubuntu – System Chrome – Application
VLC – Application Driver – System
Calculator – Application OS – System

Experiment 1 – Quick Recap





Experiment 2

File and Folder Management in a Linux GUI

Why Do We Need File Organization?

Think About This

What happens if 500 papers are kept on a table without any arrangement?

- Difficult to find information
- Files may get misplaced
- Time is wasted

Computer Equivalent

Files and folders help us keep digital information organized.

What is Linux?

Definition

Linux is an open-source operating system that manages computer hardware and software resources.

- Supports Graphical User Interface
- Supports command-line interface
- Supports multitasking
- Supports multiple users

What is GUI?

GUI = Graphical User Interface

A GUI allows users to interact with a computer using windows, icons, menus and a pointer.

Icons + Windows + Menus + Mouse/Pointer

Example

Using a smartphone by tapping icons and menus is a familiar example of graphical interaction.

Linux Desktop Environment

Important areas:

1. Desktop
2. Panel / Taskbar
3. Application Menu
4. File Manager
5. Home Directory
6. Trash

Today's Main Tool

File Manager

File Manager

What is File Manager?

A File Manager is a graphical application used to browse and manage files and folders.

- Create
- Open
- Rename
- Copy
- Move
- Delete

What is a File?

Definition

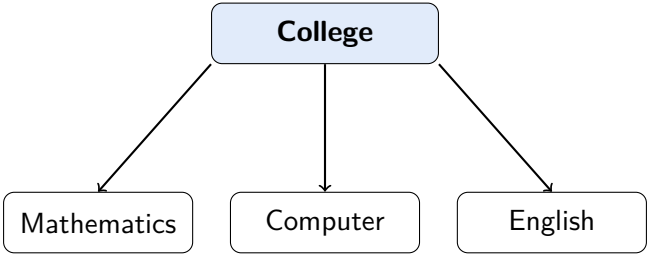
A file is a collection of information stored on a storage device.

File	Type
notes.txt	Text
photo.jpg	Image
song.mp3	Audio
video.mp4	Video
report.pdf	Document

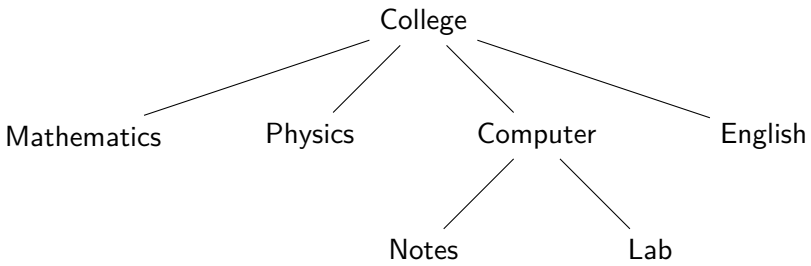
What is a Folder?

Definition

A folder is used to organize and store files and other folders.



Example: Organized Folder Structure



Good organization makes files easier to find.

Five Basic File Operations

Create | Rename | Copy | Move | Delete

Create $\rightarrow$ **Rename** $\rightarrow$ **Copy** $\rightarrow$ **Move** $\rightarrow$ **Delete**

1. Create

Meaning

Creating a new file or folder.

Example:

1. Open File Manager
2. Go to Home
3. Right-click empty space
4. Select New Folder
5. Enter the folder name

Example Folder

`ComputerLab`

2. Rename

Meaning

Changing the name of an existing file or folder.

`Student1.txt`

⇓

`StudentData.txt`

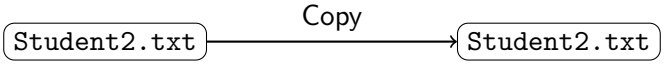
Remember

Rename changes the name; it does not create a new file.

3. Copy

Meaning

Copy creates a duplicate of a file or folder.



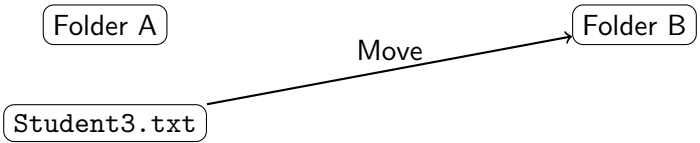
Analogy

Photocopying a document: the original remains and a copy is created.

4. Move

Meaning

Move changes the location of a file or folder.



Important

After moving, the file is no longer in its original location.

Copy vs Move

Copy	**Move**
Creates a duplicate	Changes location
Original remains	Original location becomes empty
Like photocopying	Like shifting a document

Question

I want another copy of a photo in Documents. Should I Copy or Move?

5. Delete

Meaning

Delete removes an unwanted file or folder.

- Select the file/folder
- Choose Delete / Move to Trash
- Confirm if required

Safety Tip

Always check the file name before deleting important files.

Complete GUI Demonstration

1. Open File Manager
2. Go to Home
3. Create `ComputerLab`
4. Create three text files
5. Rename one file
6. Copy one file
7. Move one file
8. Delete an unwanted file
9. Check the final folder structure

Watch the demonstration carefully.

Activity: Organize Your College Files

Task

Create the following directory structure using Linux GUI:

```
Home
|– College
|– Mathematics
|– Physics
|– Computer
    |– Notes
|– Assignments
    |– Lab
|– English
```

Activity: Weekend Trip

Start with:

```
Home/Backpack
```

Create:

```
Backpack/
├─ Food/
├─ Tools/
```

Inside `Tools`, create:

```
flashlight.txt
map.txt
```

Weekend Trip – Operations

Perform these operations:

1. Move `map.txt` from Tools to Food.
2. Copy `flashlight.txt`.
3. Rename the copy to `spare_light.txt`.
4. Delete the Food folder.
5. List everything remaining in Backpack.

Think Before You Click

Which operation should be used at each step?

GUI and Command Line

Linux provides two common ways to interact with files:

GUI

- Icons
- Menus
- Mouse
- File Manager

Command Line

- Type commands
- Terminal
- Text-based interaction

Today's focus: Linux GUI

Optional: Linux Command Equivalents

Operation	Command
Show location	`pwd`
List contents	`ls`
Create folder	`mkdir`
Create file	`touch`
Change directory	`cd`
Copy	`cp`
Move / Rename	`mv`
Remove file	`rm`

GUI first – commands later

Final Recap

Experiment 1

- Hardware
- CPU
- RAM
- HDD / SSD
- Peripheral Ports
- System Software
- Application Software

Experiment 2

- Linux
- GUI
- Files
- Folders

Quick Viva Questions

1. What is hardware?
2. What is software?
3. What is the function of CPU?
4. What is RAM?
5. What is the difference between RAM and storage?
6. What is SSD?
7. Name two peripheral ports.
8. What is system software?
9. What is application software?
10. What is Linux?
11. What is GUI?
12. What is a file?
13. What is a folder?
14. What is the difference between Copy and Move?

Remember

Identify the hardware.

Understand the software.
Organize your files.

Experiment 1 + Experiment 2